Cold regions hydrology

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5. Frozen soil processes

Learning Objectives

After today's lecture you should be able to:

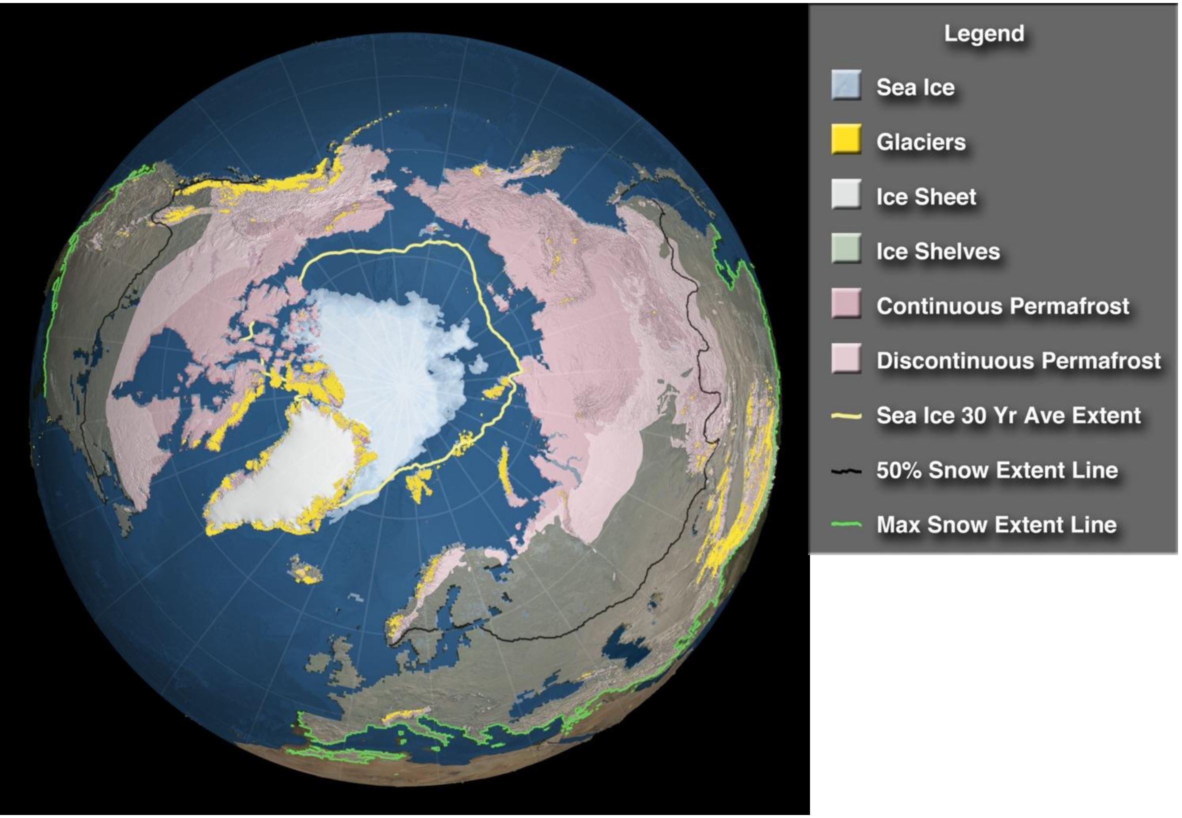
- Understand the significance and occurrence of frozen soils globally
- Understand the how snowmelt dominates the hydrological cycle in seasonally frozen soils
- Understand concepts of specific and latent heat and calculate the equilibrium state of ice and water combined in an isolated system.
- Understand the thermal properties (heat capacity and thermal conductivity) of soils
- Describe how water, ice and air are partitioned with the soil pore space
- Understand the soil freezing characteristic curve, and it's similarity to the SMC
- Appreciate the challenges of physically based modelling of frozen soils

INTRODUCTION TO FROZEN SOILS AND THEIR HYDROLOGICAL SIGNFICANCE

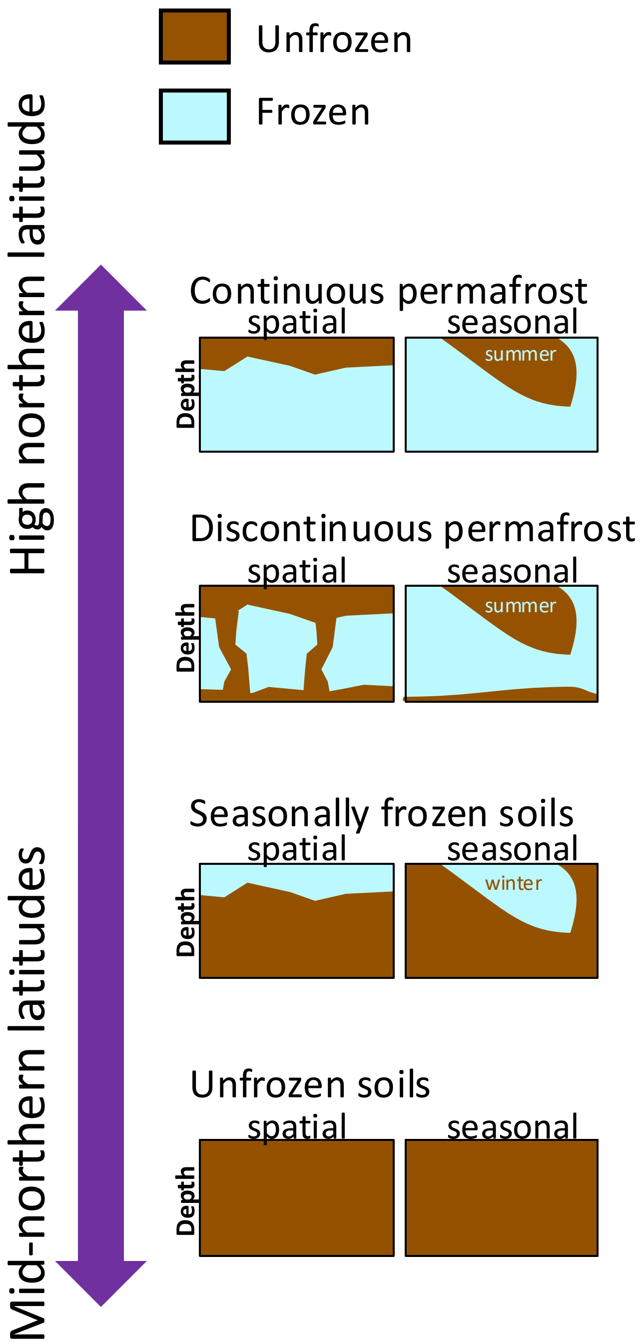
Understanding of freeze-thaw processes is fraught by "***experimental vagaries and theoretical imponderables***"

Miller (1980, Applications of soil physics)

Frozen soils - globally

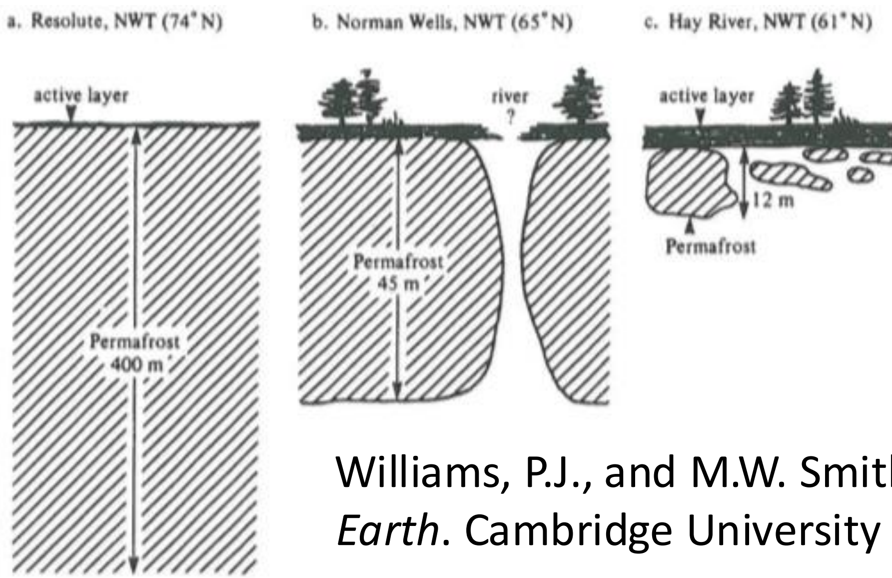


NASA/Goddard Space Flight Center
Scientific Visualization Studio
http://svs.gsfc.nasa.gov/3885



Permafrost

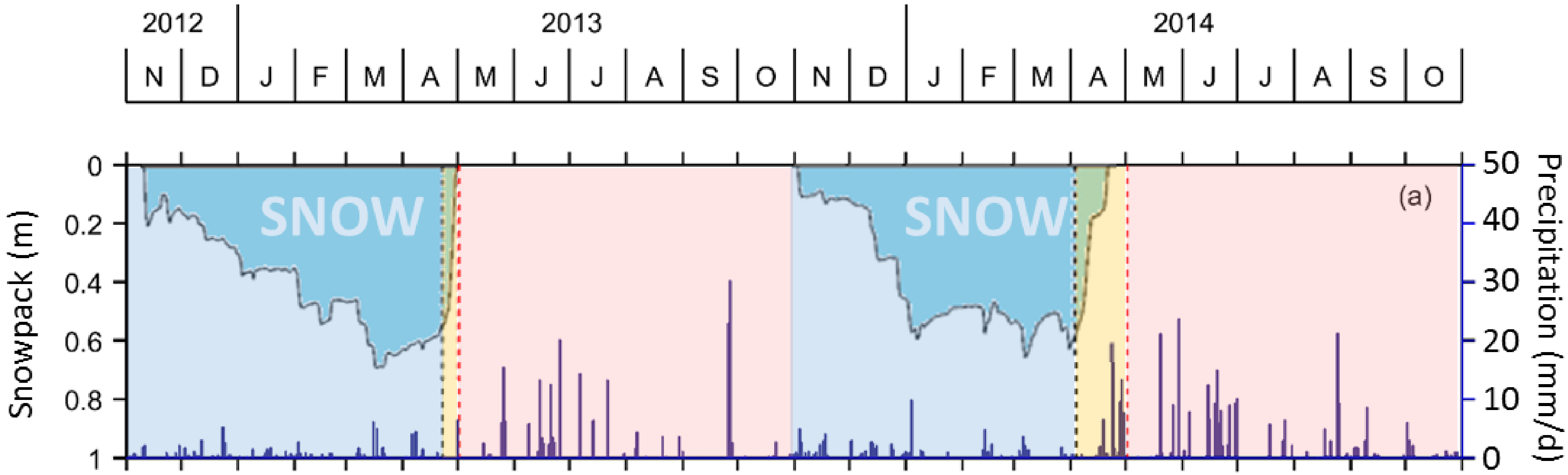
- Generally found in areas where the mean annual temperature is below -1^{o}C.
- “Continuous” vs “Discontinuous” is a spatial distinction:



Williams, P.J., and M.W. Smith. 1989. *The Frozen Earth*. Cambridge University Press.

- Permafrost thaw is associated with the release of CH_4 (major GHG)
- Permafrost thaw causes subsidence which can affect trees, roads, pipelines, etc.

Seasonally frozen soils hydrology

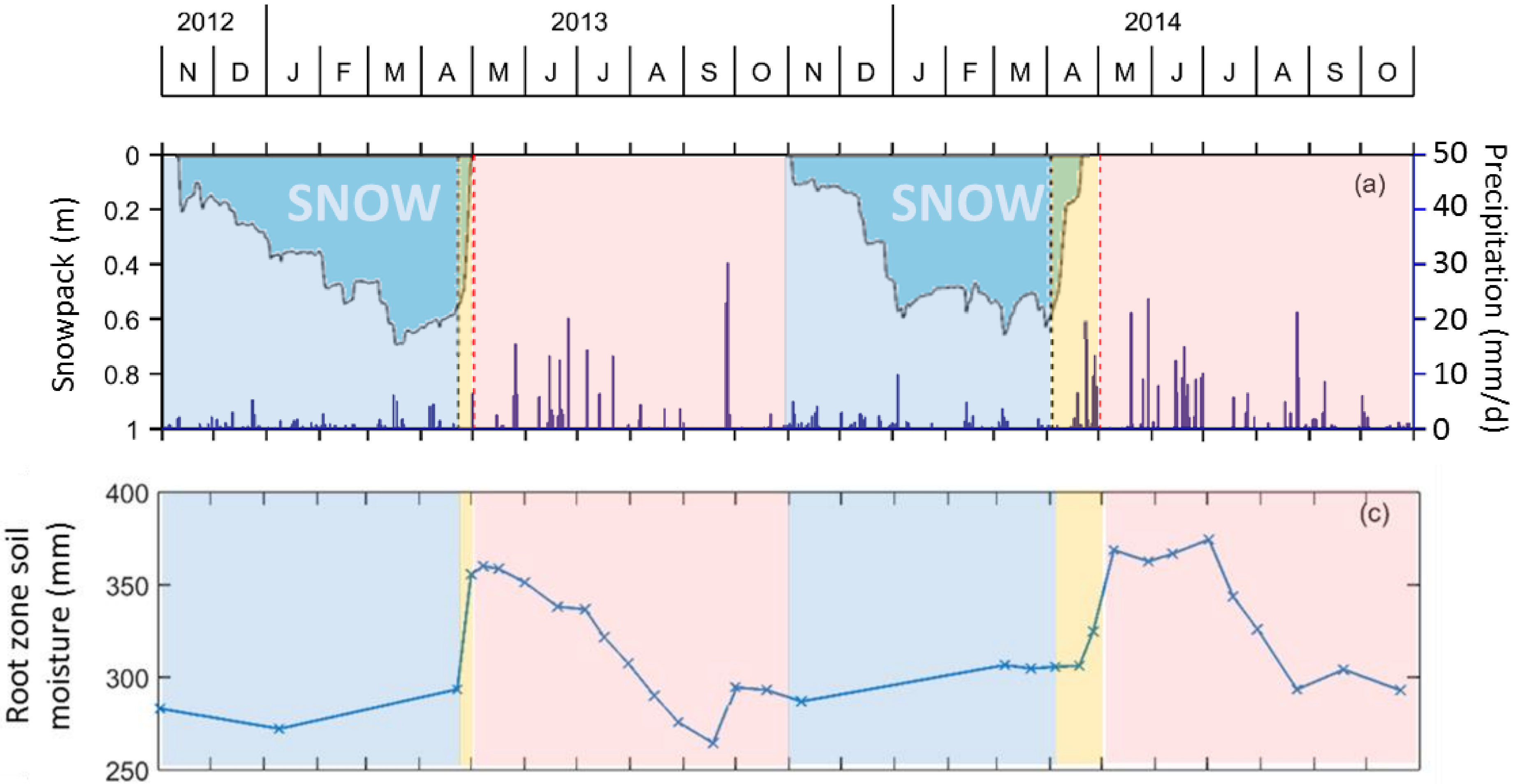


- Season 1: Snow accumulation (long and cold, slow hydrology)
- Season 2: Snowmelt (short and wet, dynamic hydrology)
- Season 3: Summer/growing season (variable)
- Northern-hemisphere “water year” typically Oct-Sept.

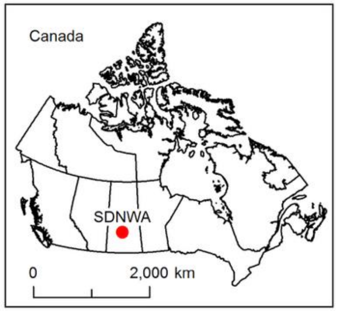
https://doi.org/10.5194/hess-21-5401-2017

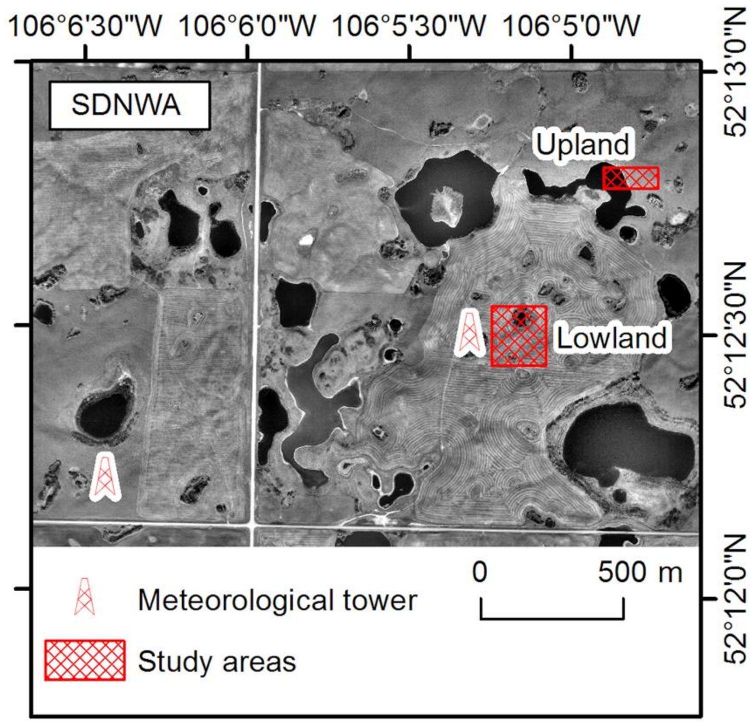
Seasonally frozen soils hydrology

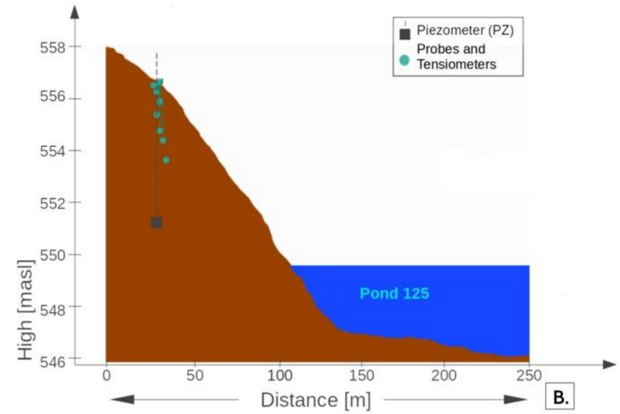
https://doi.org/10.5194/hess-21-5401-2017

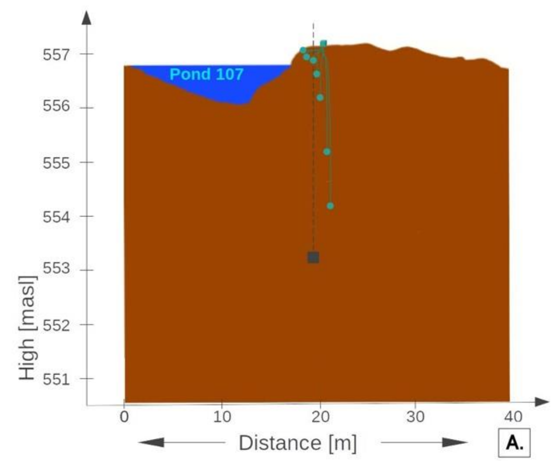


Field site – St Denis SK



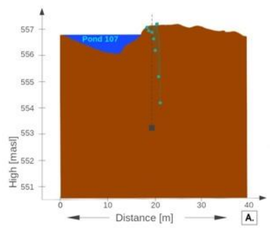


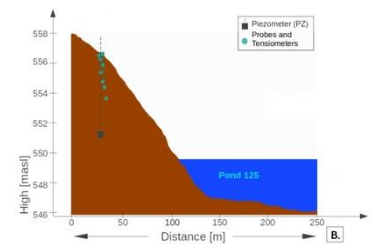




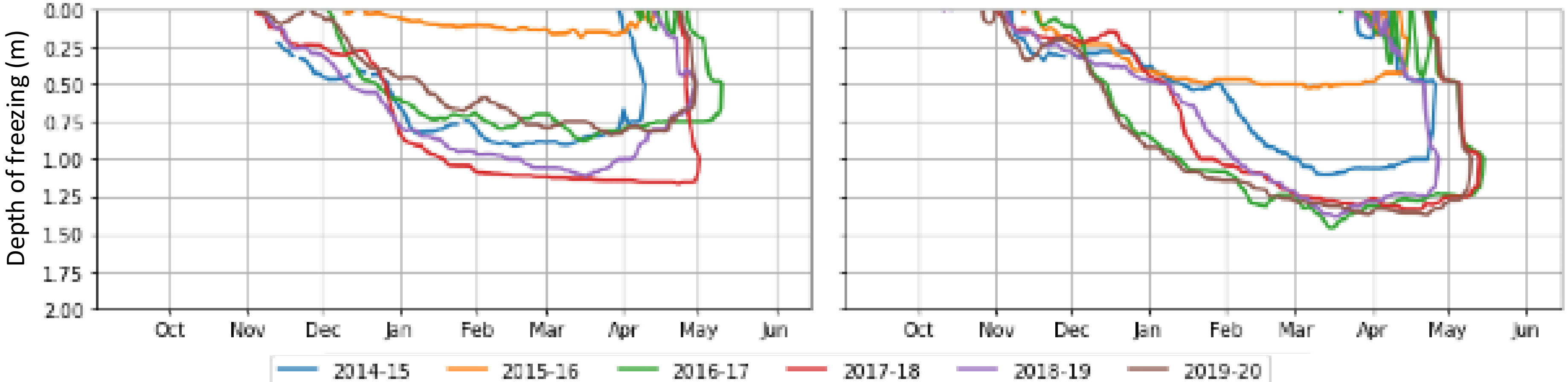
Sanchez-Rodriguez, I., Ireson, A., Brannen, R., & Brauner, H. (2025). Insights into freeze–thaw and infiltration in seasonally frozen soils from field observations. *Vadose Zone Journal*, *24*(1), e20396.
https://doi.org/10.1002/vzj2.20396

Observed soil freezing depths





LOWLAND PROFILE **UPLAND PROFILE**



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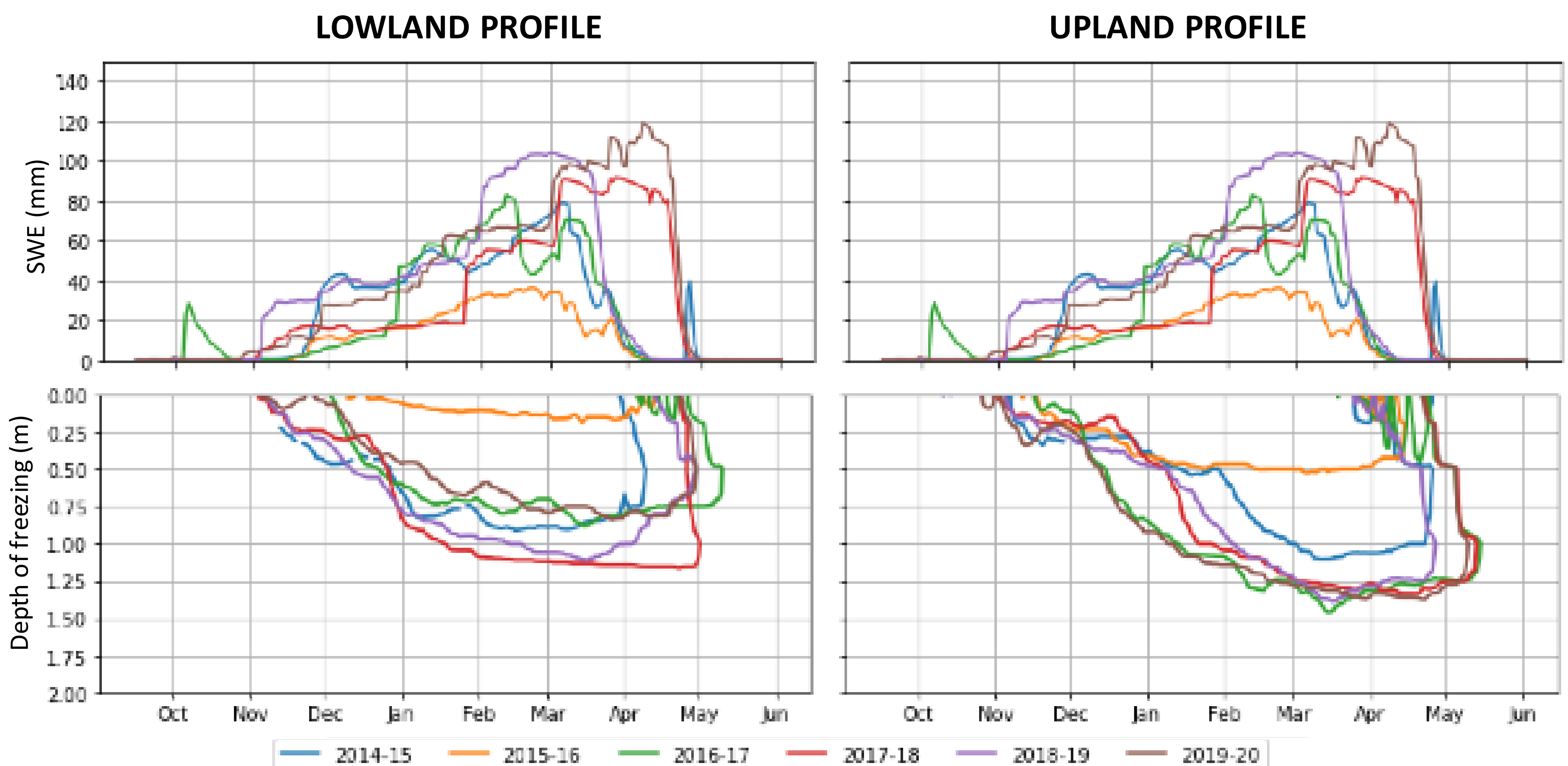

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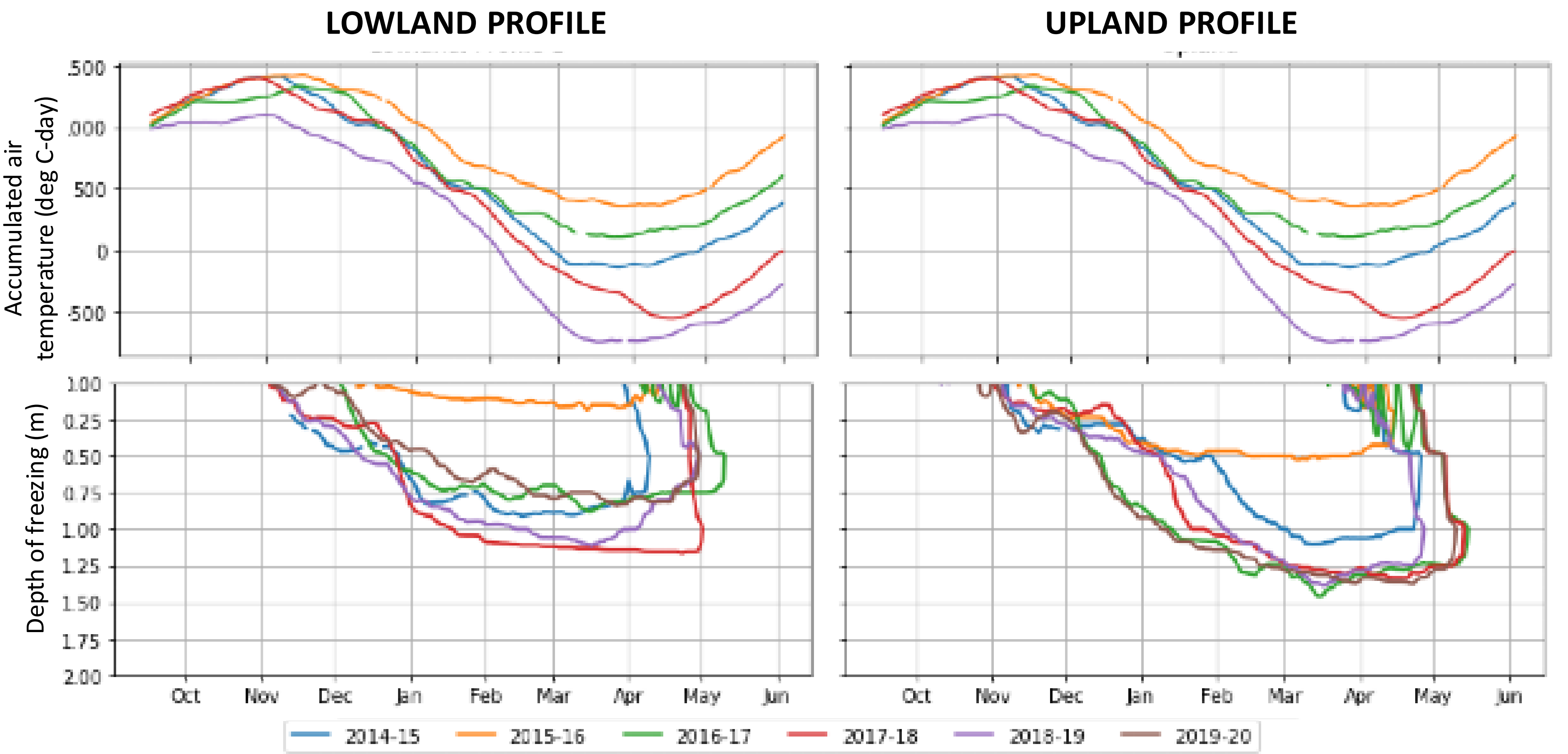

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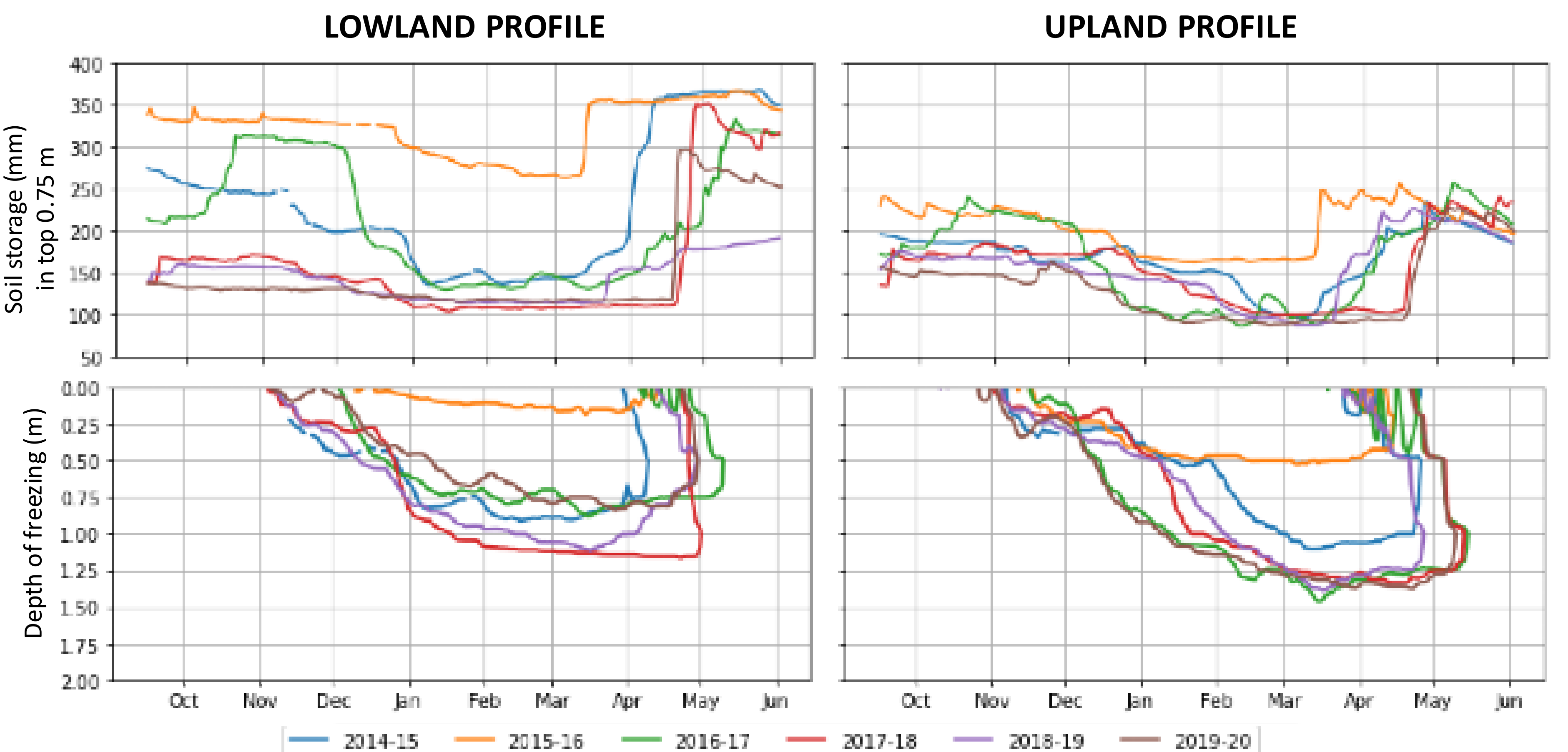

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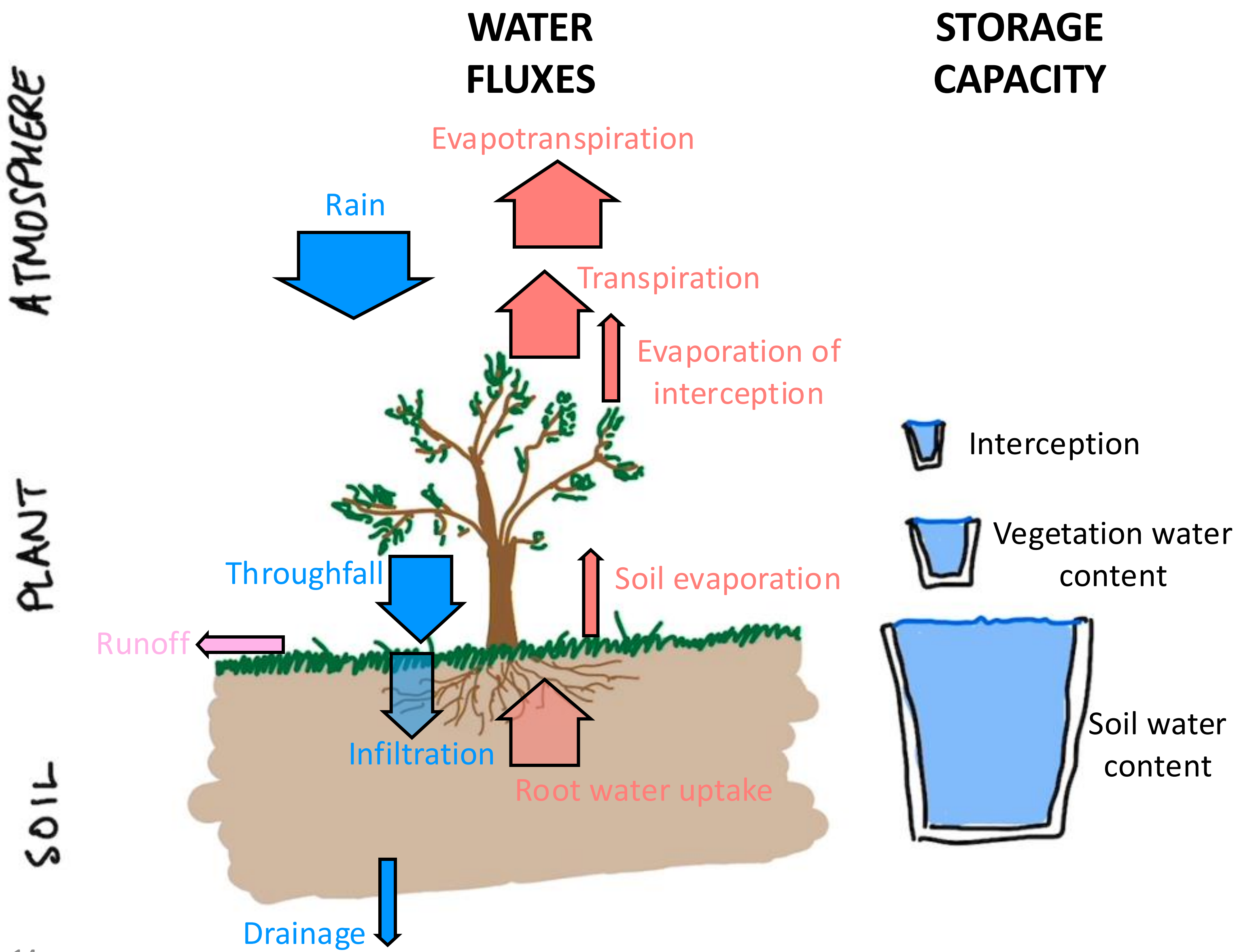
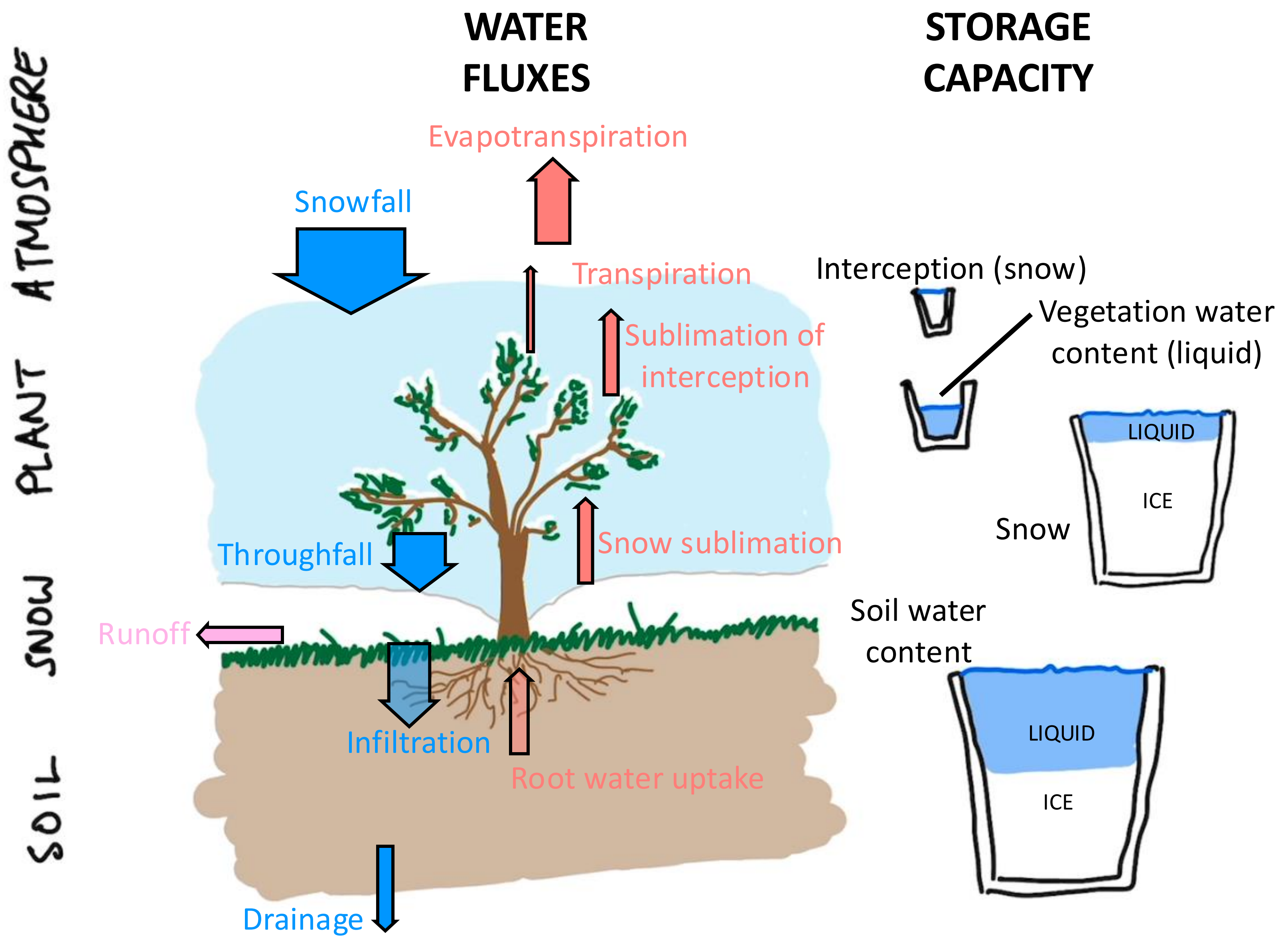
Summer soil water balance



Winter (cold region) soil water balance

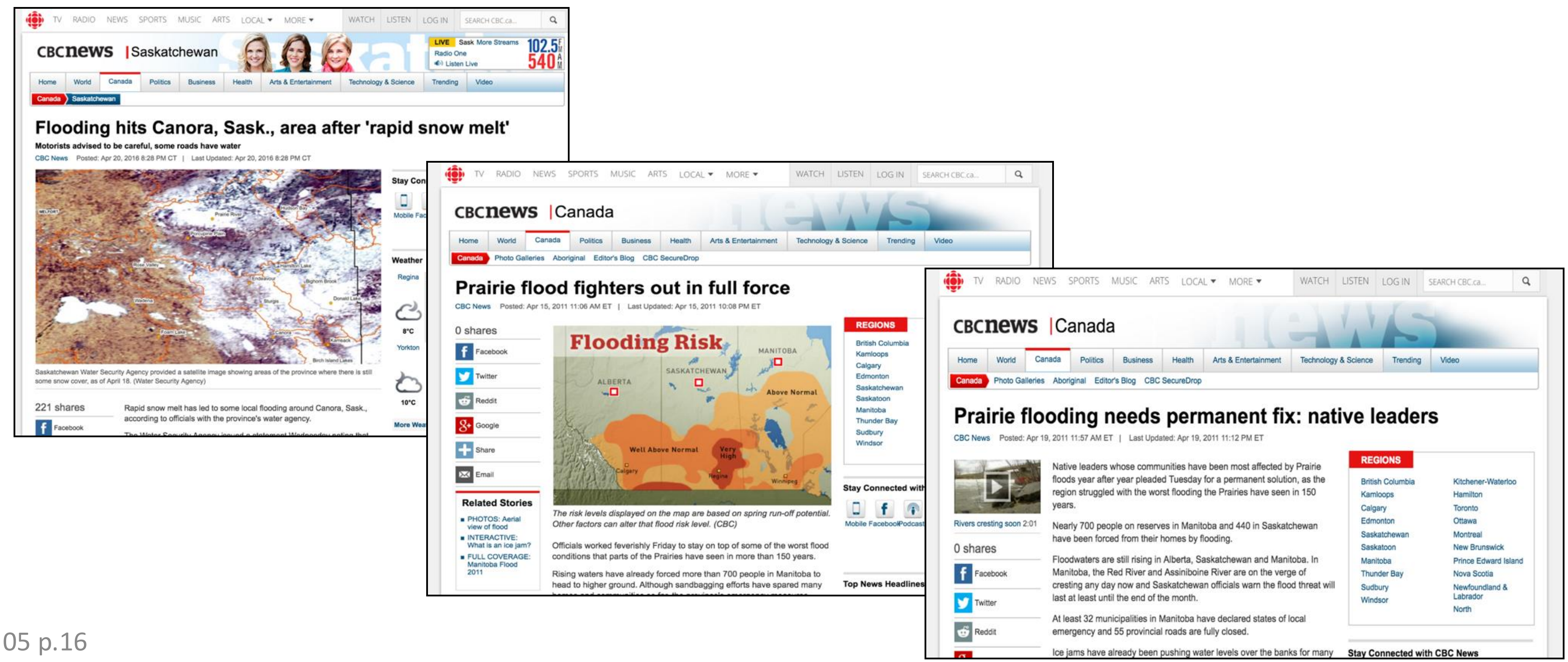


Critical question:

How is snowmelt and spring rainfall partitioned between infiltration into and runoff over frozen soils?

$$SWE + P = I + R$$

This is a key driver of flood risk in the Canadian prairies.



Empirical models for infiltration

A simple and attractive approach which is suitable for many practical problems, is to use an empirical model. One classic model is that developed by Don Gray.

Consider three classes of soils

- **Restricted**: Ice lenses in soil prevent infiltration, $I = 0$
- **Unlimited**: Air-filled macropores absorb meltwater, $I = SWE$
- **Limited**: Infiltration depends on S

	Low SWE	High SWE
Low S	Low runoff	Low runoff
High S	Moderate runoff	High runoff

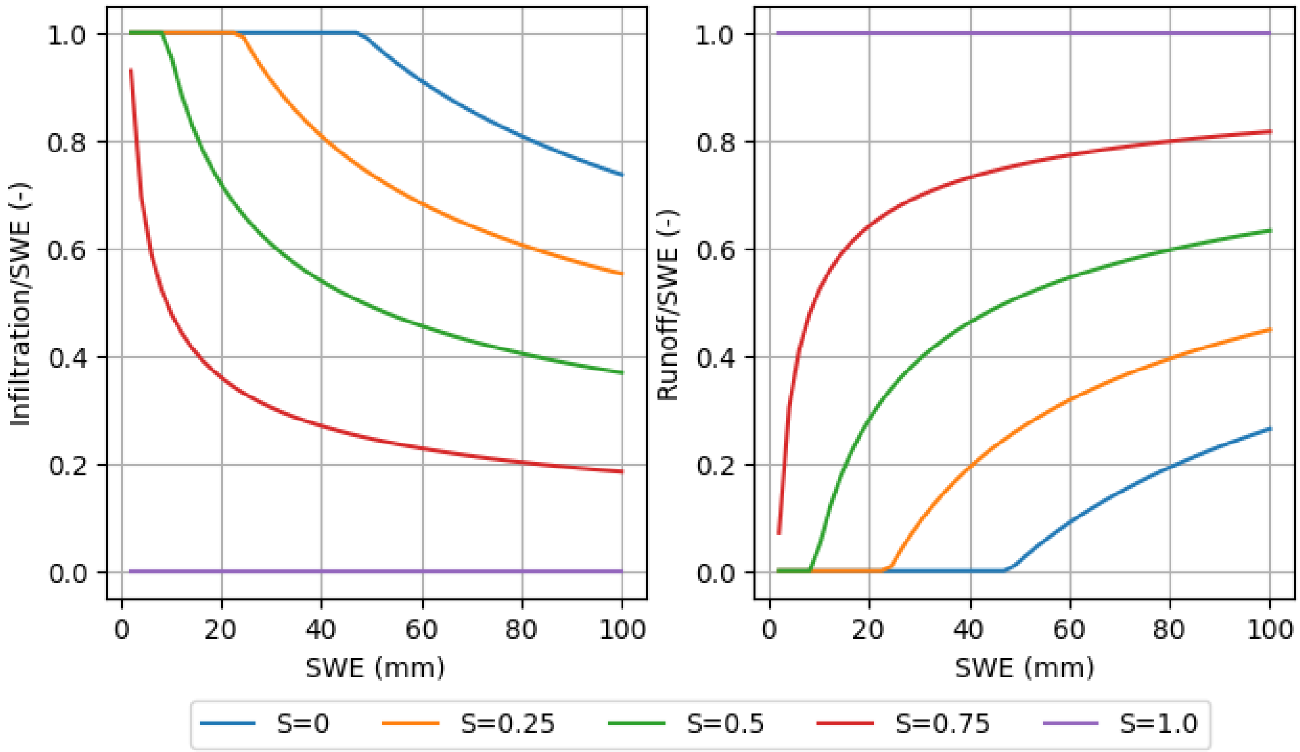
$$I = 5(1 - S)SWE^{0.584}$$

$$R = SWE - I$$

I = infiltration (mm)
R = runoff (mm)
SWE = Snow Water Equivalent (mm)
S = Soil saturation (-)

Zhao, L., & Gray, D. M. (1997). A parametric expression for estimating infiltration into frozen soils. *Hydrological Processes*, *11*, 1761–1775.

Empirical models for infiltration



All snow runs off when $S = 1.0$ (i.e. zero infiltration)

Most snow infiltrates when $S = 0.0$ (but depends on SWE – high SWE still creates some runoff)

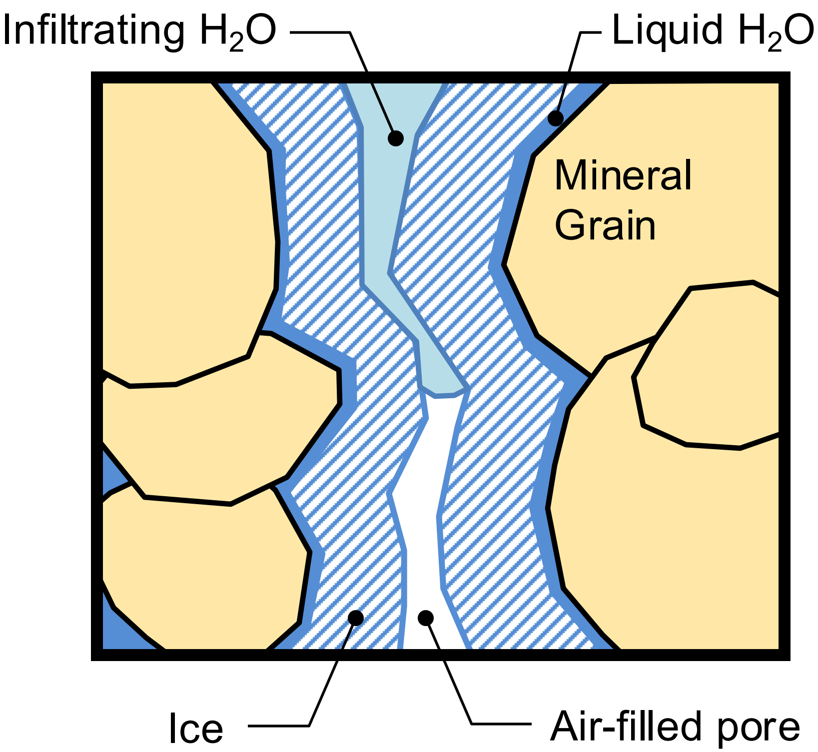
Mix of runoff and infiltration when $0 < S < 1$ (depending on SWE)

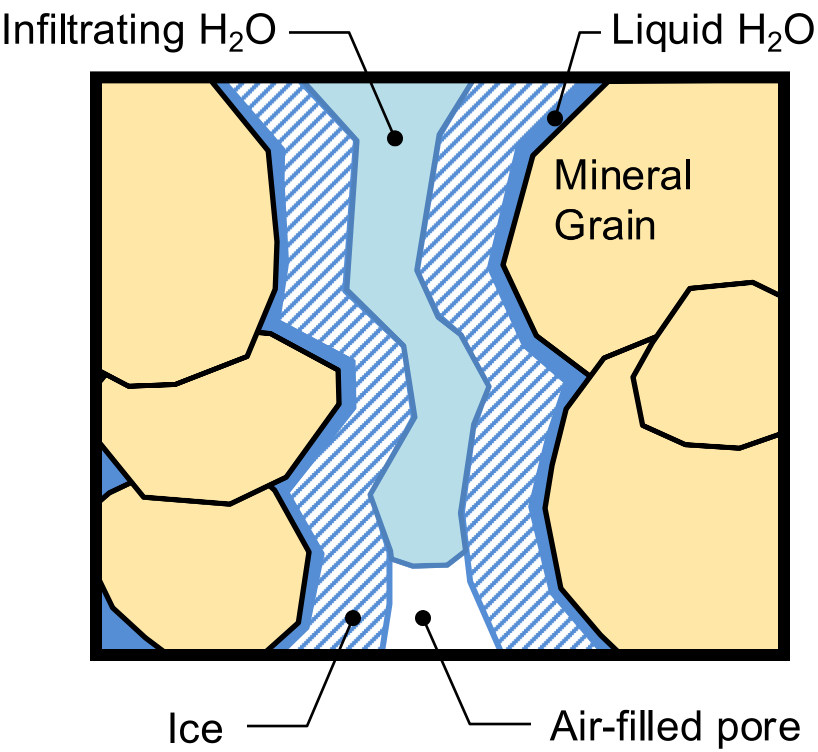
Infiltration into frozen soils

In frozen soils, the pore space contains some combination of ice, liquid water, and air, and this impacts on the hydraulic properties, and in particular the $K(\psi)$ relationship.

Infiltration and percolation are thus impeded by the presence of ice.

However, it is easy to ***over-estimate*** this impedance, if the role of air-filled macropores is overlooked





Images by Charles Maule, 2019

Physically based models for infiltration

Richards' Equation describes the movement of water though unsaturated porous media, and can simulate infiltration given an appropriate upper boundary condition

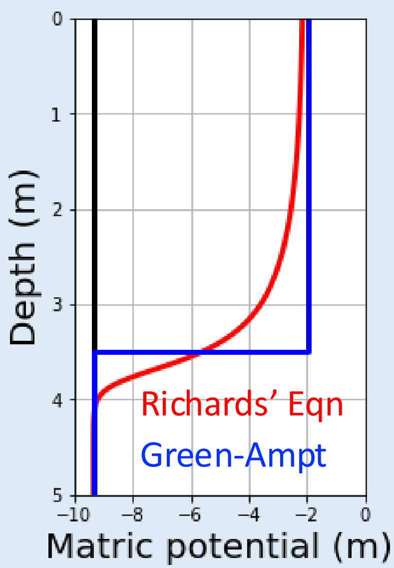
$$C(\psi)\frac{\partial\psi}{\partial t} = \frac{\partial}{\partial z}\left(K(\psi)\left(\frac{\partial\psi}{\partial z} + 1\right)\right)$$

However, this is onerous to solve, so various simplifications have been proposed:

Green and Ampt method

Infiltration is treated as a square-wave propagating with depth.

Analytical solution procedure for I is well described by Dingman, p. 366



Phillip equation

This is an analytical solution to the upper boundary of Richards' Equation, subject to a number of simplifying assumptions.

Infiltration is treated as a diffusive process (which is not quite right, but close) and gravitational potential is ignored.

Method is described in Dingman, p. 361

Physically based models for frozen soils

The Green and Ampt and Phillip equations can be adapted for infiltration in frozen soils, by accounting for the role of ice in the pore space.

However, these are limited methods (even in unfrozen soils), so a true physically based methods require numerical solutions to the PDEs.

We therefore need to solve the **coupled mass** (i.e. water flow, based on Richards' Equation) and **heat transport equations**

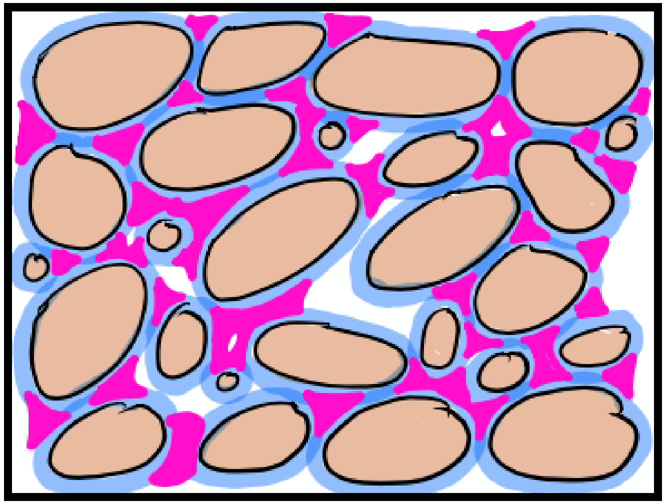
$$\frac{\partial \theta_l(\psi)}{\partial t} + \frac{\rho_i}{\rho_l}\frac{\partial \theta_i(T)}{\partial t} = \frac{\partial}{\partial z}\left[K(\psi)\left(\frac{\partial \psi}{\partial z} + 1\right)\right] + \frac{1}{\rho_l}\frac{\partial q_v}{\partial z} - S$$

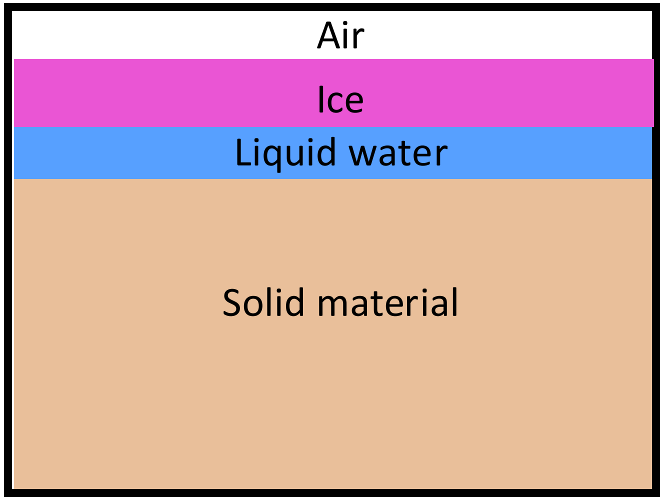
$$C\frac{\partial T}{\partial t} - L_f \rho_i \frac{\partial \theta_i}{\partial t} = \frac{\partial}{\partial z}\left[\kappa_T \frac{\partial T}{\partial z}\right] - \rho_l c_l \frac{\partial q_l T}{\partial z}$$

For the remainder of this lecture we will discuss the physics behind these equations.

Soil liquid and ice content

In frozen soils, the air space is partitioned between air, liquid water and ice.





We still have the volumetric liquid water content: $\theta_L = V_L/V_T$

We now also have the volumetric ice content: $\theta_I = V_I/V_T$

However, the density of ice, $\rho_I = 918$ kg/m³ is less than that of liquid water, $\rho_L = 1000$ kg/m³, meaning that ice actually expands. Therefore the volume of equivalent liquid water in the ice phase, i.e. the ice-water equivalent (like SWE) is given by

$$\frac{V_{IWE}}{V_T} = \frac{\theta_I \rho_i}{\rho_L} \approx 0.918\theta_I$$

Volumetric heat capacity

We have seen that the heat capacity of a single substance is given by

$$\Delta U = cm\Delta T$$

With units:

$$\mathrm{J} = \frac{\mathrm{J}}{\mathrm{kg.K}}.\mathrm{kg.K}$$

It is convenient to express this on a per unit volume basis, i.e.

$$\frac{\Delta U}{V} = c\frac{m}{V}\Delta T = c\rho\Delta T$$

And rearranging

$$\Delta U = cV\rho\Delta T = c_V V\Delta T$$

Where $c_V = c\rho$ is the volumetric heat capacity, with units J/m^3/K (check this).

Volumetric heat capacity of soil

The volumetric heat capacity of a single substance is

$$c_V = \frac{\Delta U}{V \Delta T}$$

Since frozen soil involves four components, the bulk volumetric heat capacity, c_{VB} of some control volume of soil, V_T, is the volume weighted sum of all the component heat capacities, i.e.

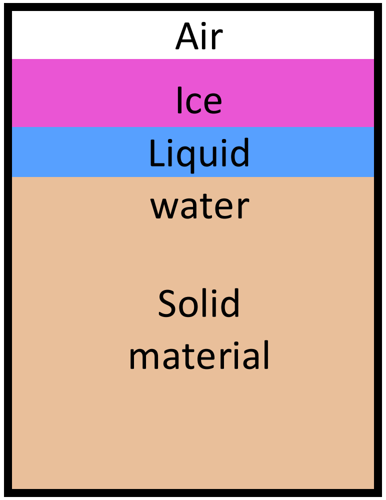
$$c_{VB} V_T = c_{VL} V_L + c_{VI} V_I + c_{VA} V_A + c_{VS} V_S$$

Hence

$$c_{VB} = c_{VL}\theta_L + c_{VI}\theta_I + c_{VA}(n - \theta_L - \theta_I) + c_{VS} n$$

Where n is again the porosity (-). Typical parameter values are

	ρ (kg/m^3)	c (J/kg/K)	c_V (MJ/m^3/K)
Liquid	1000	4180	4.18
Ice	918	2100	1.93
Air	1.25	1006	0.001
Solids	2650	755	2.00



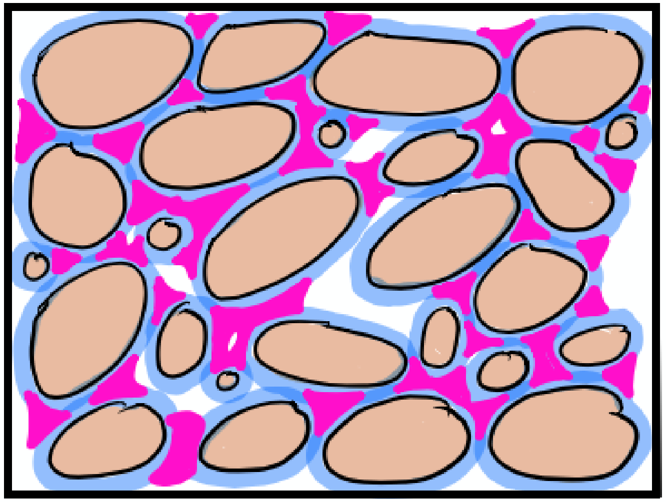
Thermal conductivity

The thermal conductivity of a substance determines how rapidly heat can be conducted through that substance, and is the constant of proportionality in Fourier's law:

$$J_H = -\kappa \frac{dT}{dx}$$

Where J_H is the heat flux (J/m^2/s), κ is the thermal conductivity (J/K/m/s or W/K/m), T is temperature (K) and x is distance (m).

The bulk thermal conductivity depends not only on the composition of the soil, but also the orientation of the different components – it is all about the connected heat flow paths.



Thus it is more complex to calculate than the bulk heat capacity. Classical methods to do this include Johansen (1975) and Farouki (1981):

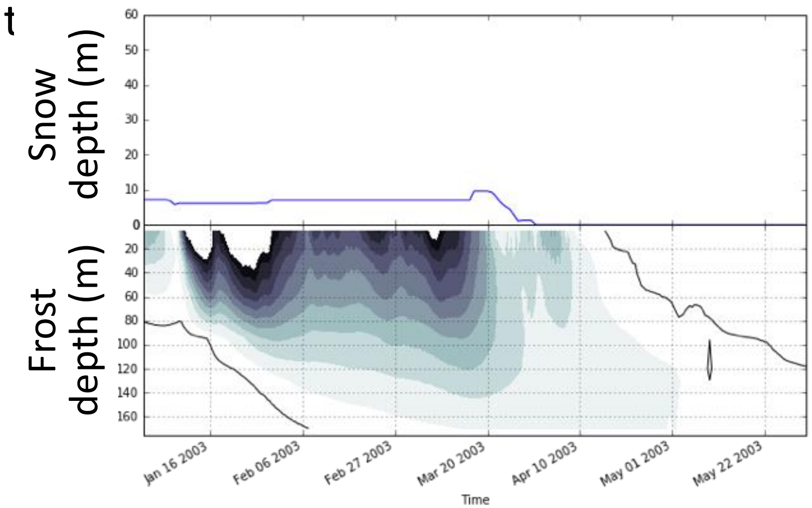
Johansen, O.: Thermal conductivity of soils, Ph.D. thesis, Ph.D. dissertation, Norwegian Technical University, Trondheim, 1975.
Farouki, O. T.: The thermal properties of soils in cold regions, Cold Reg. Sci. Technol., 5, 67–75, 1981.

Snow insulation

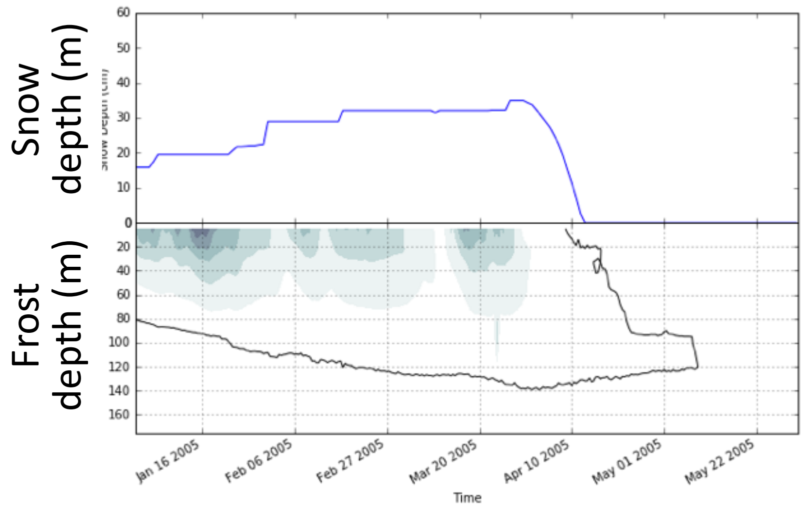
Snow is an excellent insulator, which means it has a relatively low thermal conductivity

More snow → less frozen ground

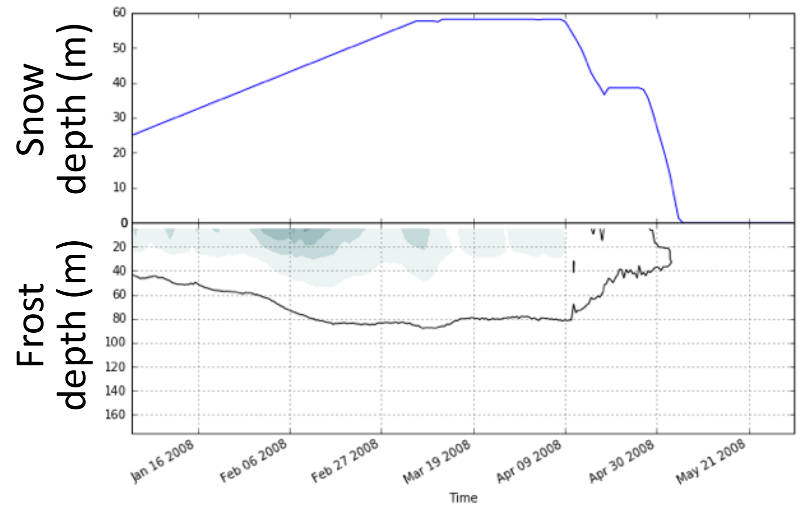
2003



2005



2008



Free water vs soil pore water

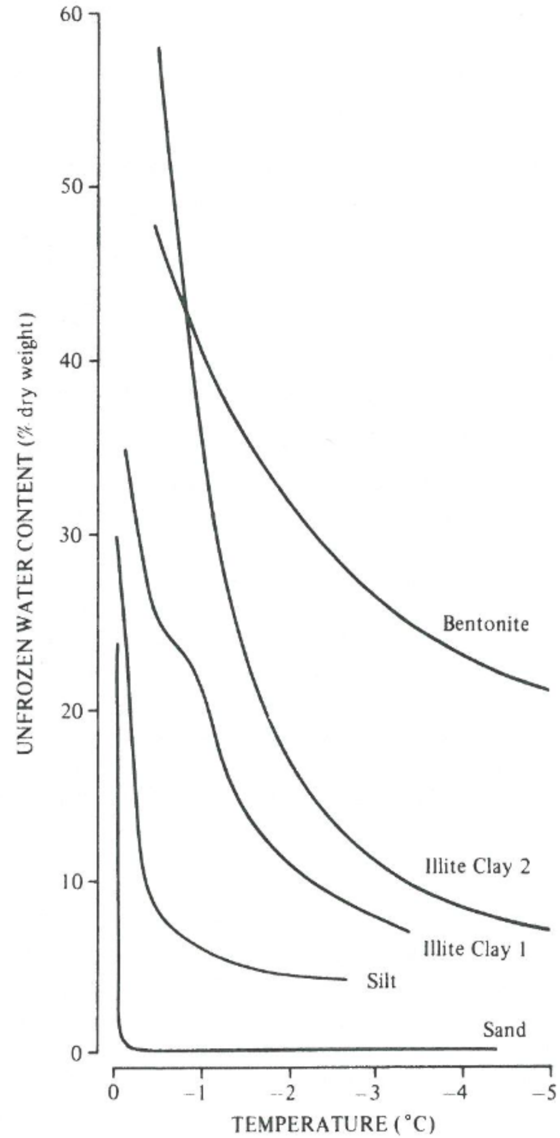
In a free water body, internal energy is either accumulated as specific heat or latent heat, but not both at the same time. Water freezes at 0°C, and there is no change in temperature while phase change is occurring.

This is not true of soils, due to the phenomenon of **freezing point depression**.

For some reason (to be discussed), the water in soil pores freezes over a range of temperatures below 0°C

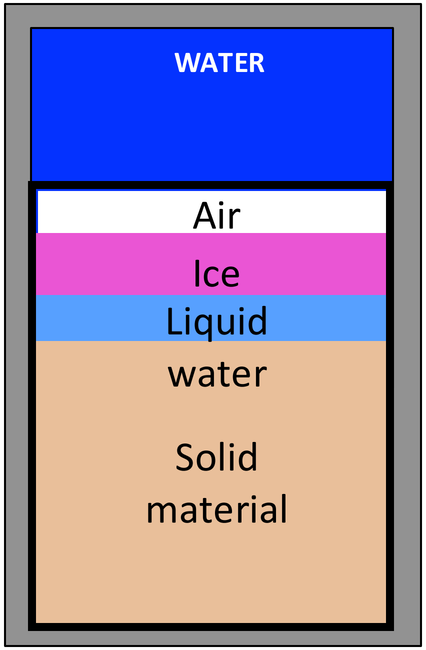
As a result, some pore remain liquid filled at sub-zero temperatures.

Latent heat is absorbed/released more gradually, over a range of temperature.



Williams, P. J., & Smith, M. W. (1989). *The Frozen Earth*. Cambridge University Press.

Soil ruins everything!



Imagine that I present you with the same problem we had earlier, but instead of just water and ice, we now have a soil system… i.e.

Imagine that we have 0.5 kg of soil at -4°C with $\theta_L = 0.05$, $\theta_I = 0.1$, $n = 0.4$ and all properties given below. What happens when we add 0.5 kg of liquid water at +1°C?

Due to freezing point depression, we cannot separate out the specific and latent heat calculations – we have changes in internal energy due to both occurring simultaneously!

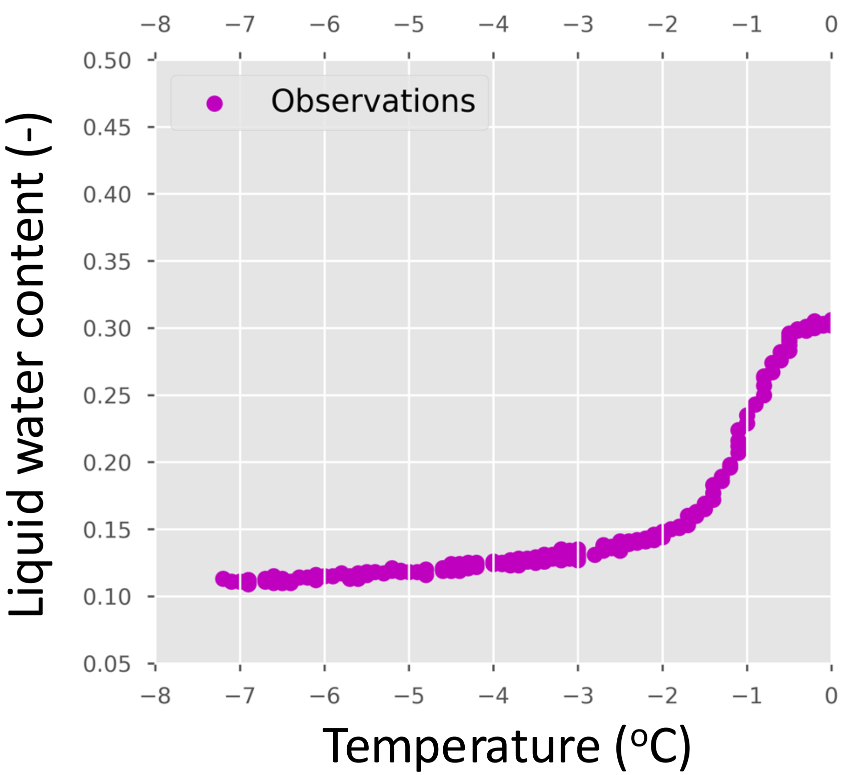
This makes the solution extremely difficult… basically you need a model to do this correctly.

A simplified solution to this problem can be achieved by ignoring the effects of freezing point depression (i.e. assuming that all the phase change takes place at 0°C).

(Note, you might need to refer to this slide for your assignment)

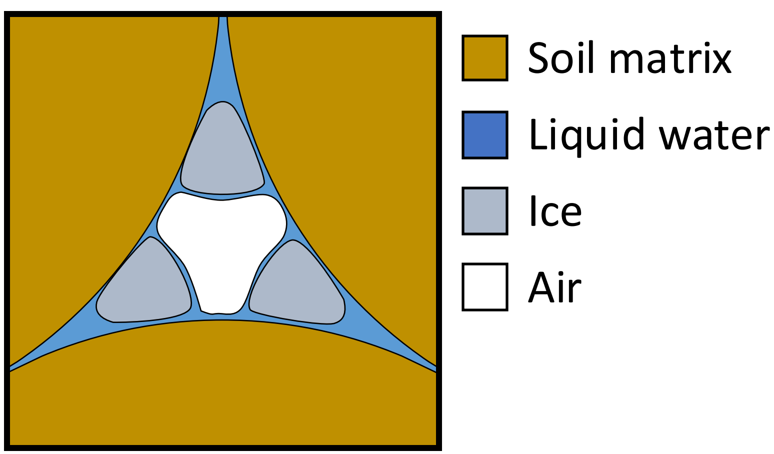
Freezing point depression

Freezing point depression is certainly a real phenomenon and can be readily observed in the field, e.g. these field data are from St Denis, SK



The simple version of the conventional explanation for this phenomenon is that the capillary and adsorption forces provide additional "free-energy" to the pore water, which makes phase change more expensive (more energy must be removed to freeze the soil/less energy must be added to thaw the soil).

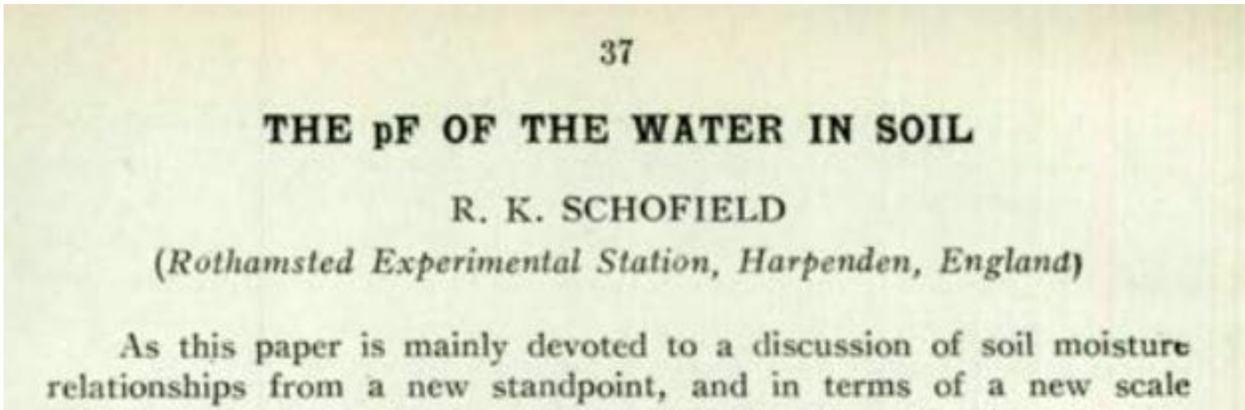
Hence, water, ice and air co-exist in the pore space:



After Miller (1980, Applications of soil physics)

The Generalized Clapeyron Equation

The earliest mention of the phenomenon of freezing point depression in soils goes back to a monograph by Schofield in 1935:



37

THE pF OF THE WATER IN SOIL

R. K. SCHOFIELD

(Rothamsted Experimental Station, Harpenden, England)

As this paper is mainly devoted to a discussion of soil moisture relationships from a new standpoint, and in terms of a new scale

Schofield, R. K. (1935). The pF of the water in soil. *Third International Congress of Soil Science*, *2*, 37–48.

Schofield had no interest in actual frozen soils, but was interested in freezing soils in a laboratory to determine their soil moisture characteristic curve a low suctions (where e.g. the hanging column experiment would fail due to cavitation).

A version of the mathematical model that Schofield proposed came to be known as the Generalized Clapeyron equation, and it states:

$$\frac{dP}{dT} = \frac{L_f \rho_L}{T}$$

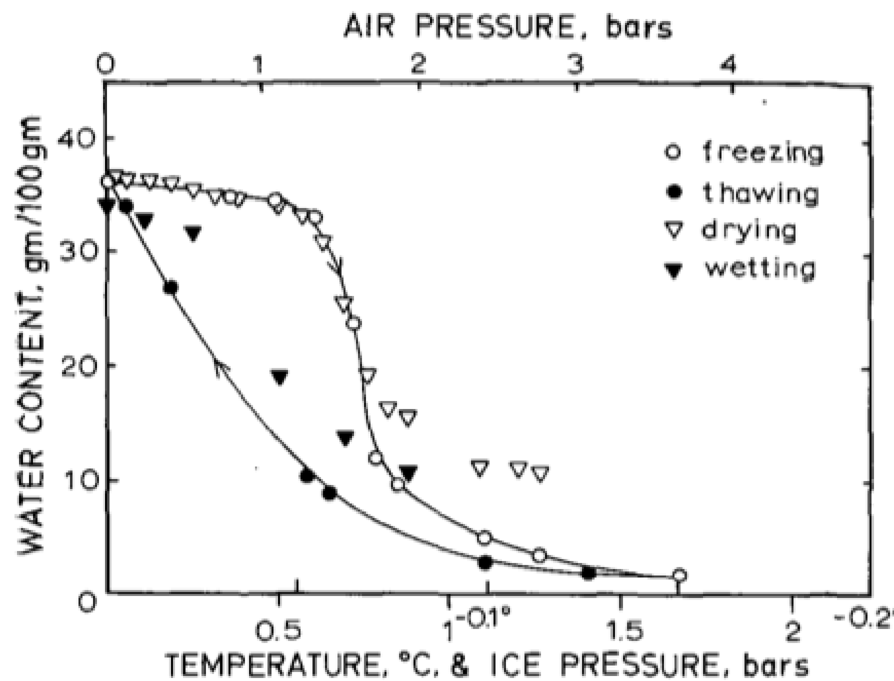
That is, the change in equilibrium pressure (and hence matric potential) at which phase change occurs is a function of temperature.

Soil freezing characteristic curve

The GCE results in a linear relationship between freezing temperature and matric potential, such that a temperature of -1°C exerts an equivalent matric potential of -122 m.

Recall that the Soil Moisture Characteristic (SMC) is the relationship between water content and matric potential and is applied in unfrozen soils.

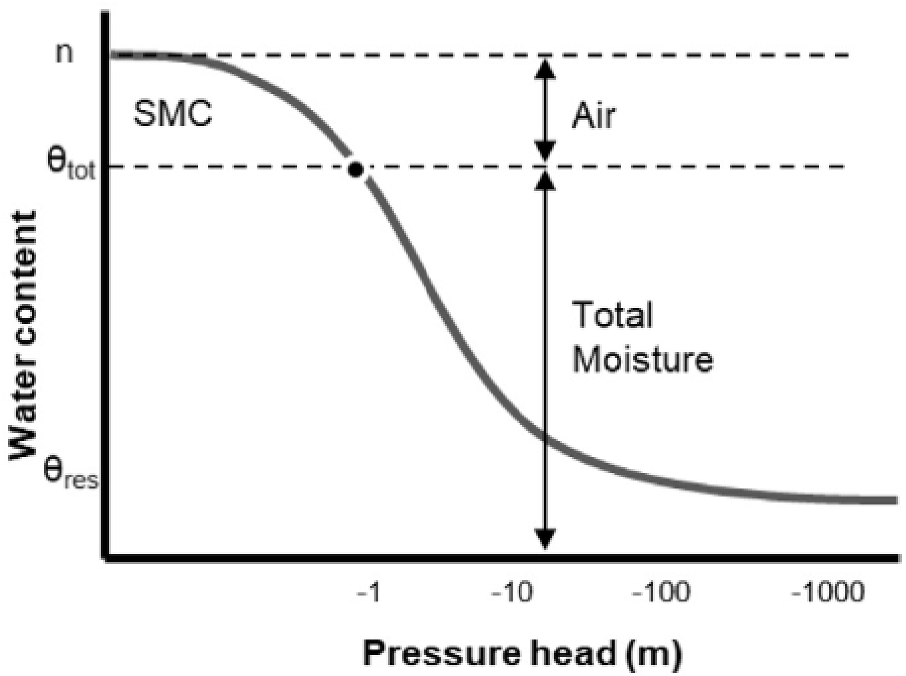
In frozen soils, the GCE is applied to produce a Soil Freezing Characteristic (SFC), which is the relationship between liquid water content and temperature.

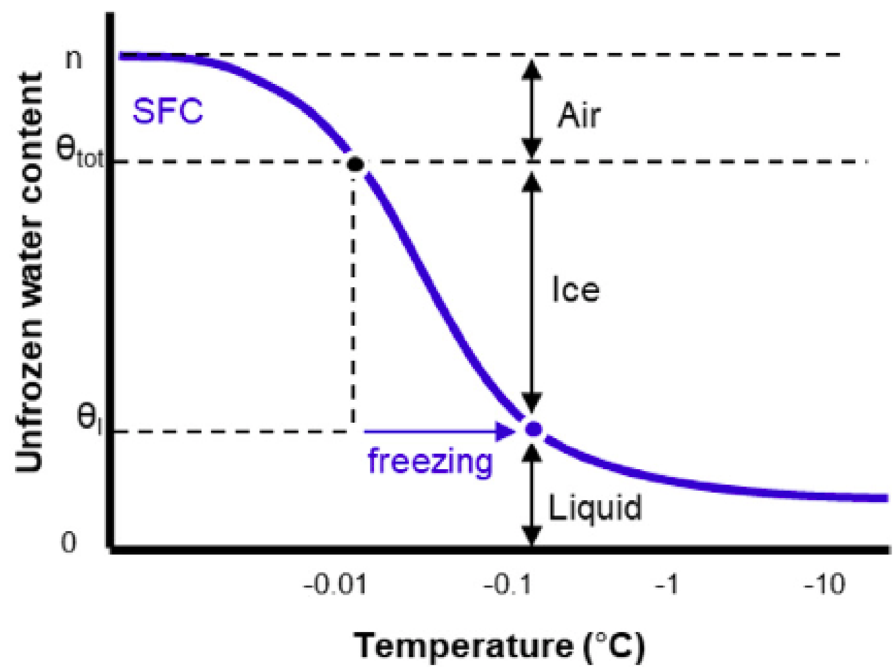


Koopmans, R. W. R., & Miller, R. D. (1966). Soil freezing and soil water characteristic curves. *Soil Science Society of America Journal*, *30*(6), 680–685.

Soil freezing characteristic curve

Whereas water is replaced by air during drying with the SMC, liquid water is replaced by ice during freezing in the SFC.

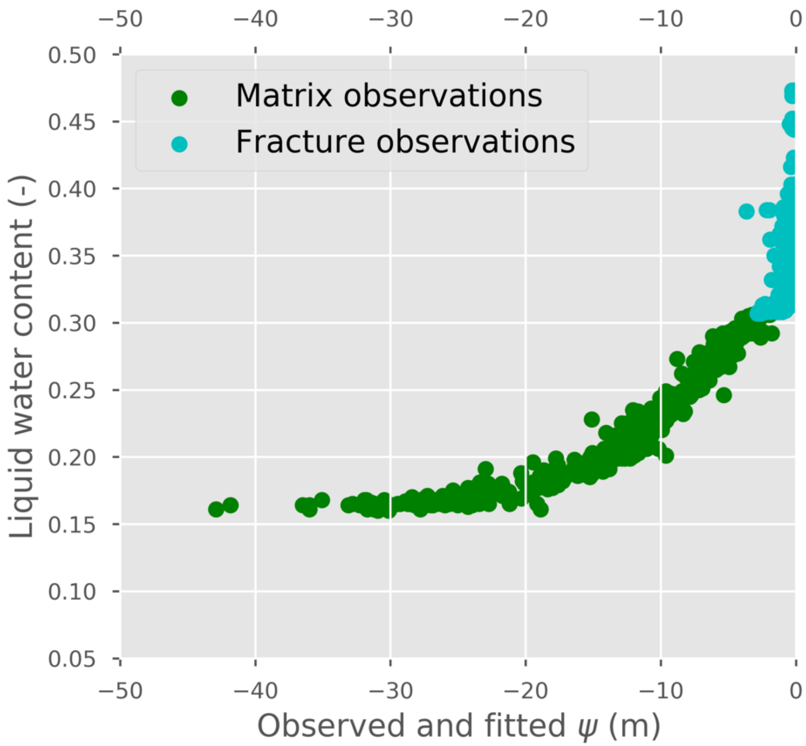


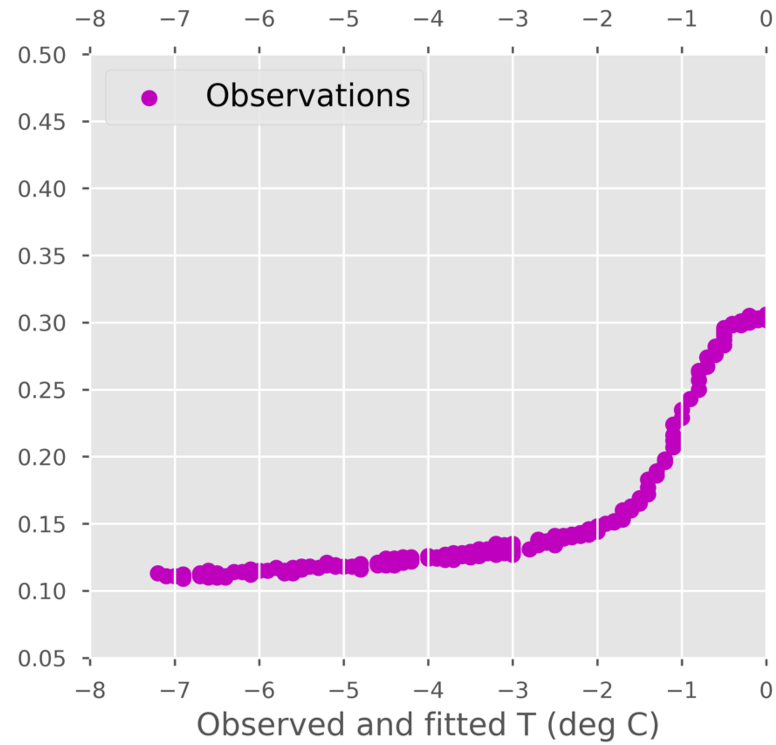


Mohammed, A. A., Kurylyk, B. L., Cey, E. E., & Hayashi, M. (2018). Snowmelt Infiltration and Macropore Flow in Frozen Soils: Overview, Knowledge Gaps, and a Conceptual Framework. *Vadose Zone Journal*, *17*(1), 1–15. https://doi.org/10.2136/vzj2018.04.0084

Soil freezing characteristic curve

Such curves can be readily observed in the field: e.g. these data are from St Denis, SK.

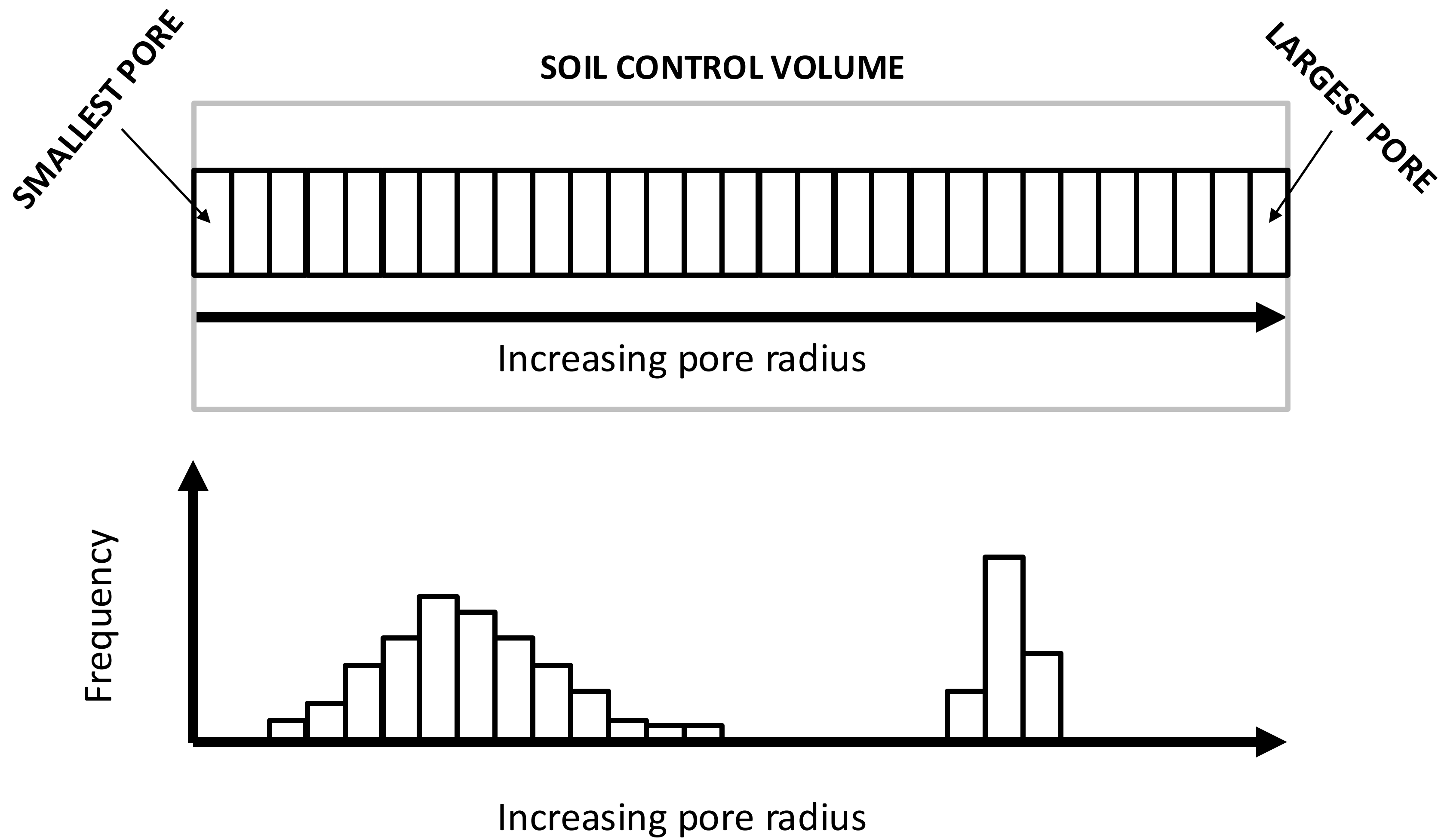




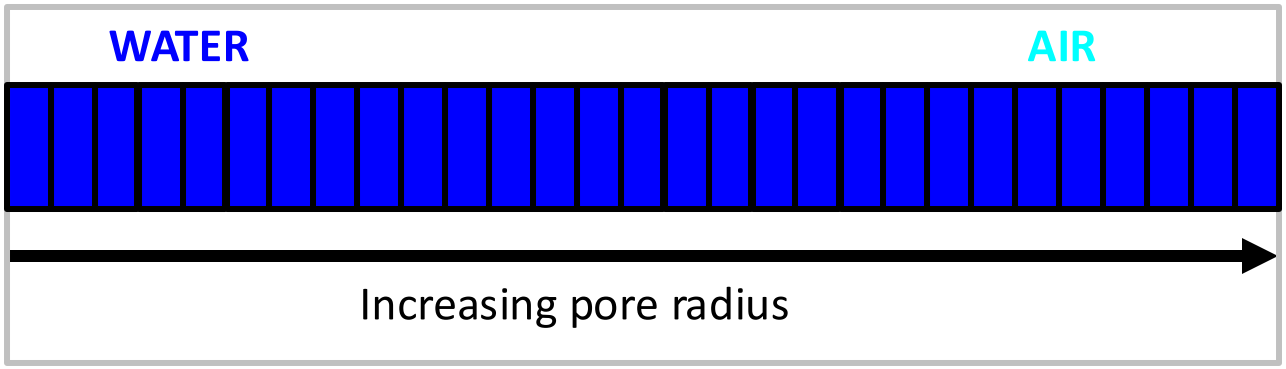
Drying curve data:
Summer 2017
5cm depth

Freezing curve data:
Winter 2016/17
5cm depth

Conceptual model of freezing soils

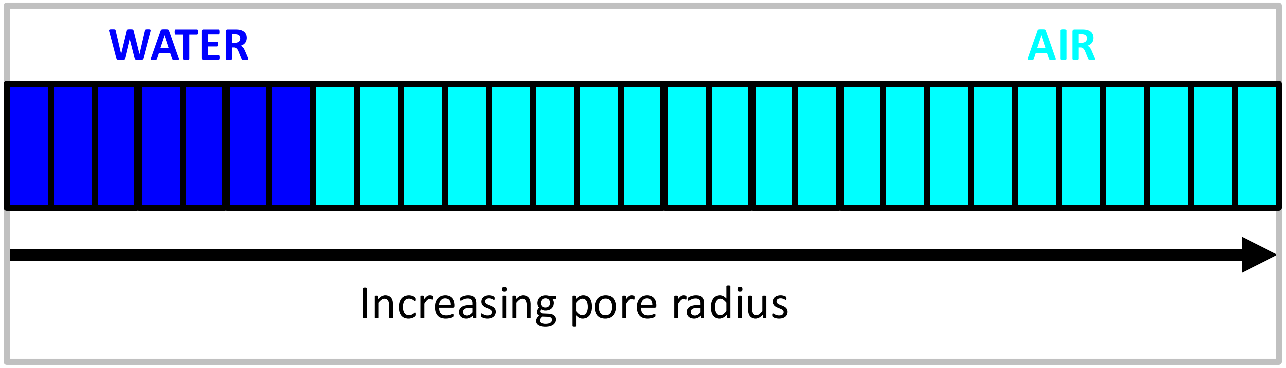


Conceptual model of wetting/drying



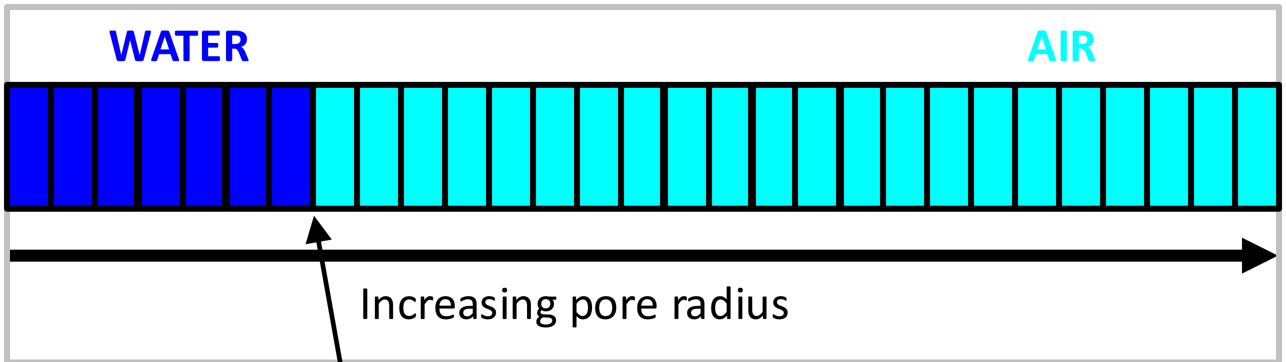
WETTING

Conceptual model of wetting/drying



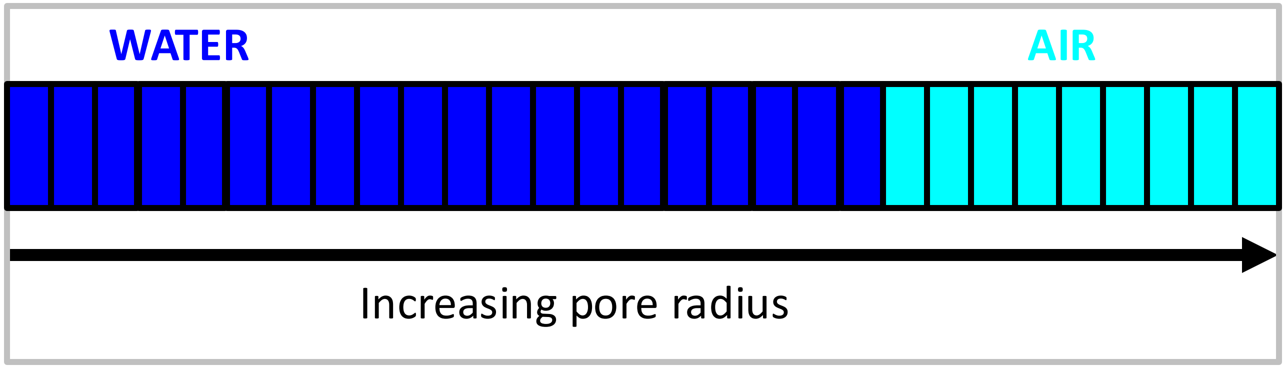
DRYING

Conceptual model of wetting/drying



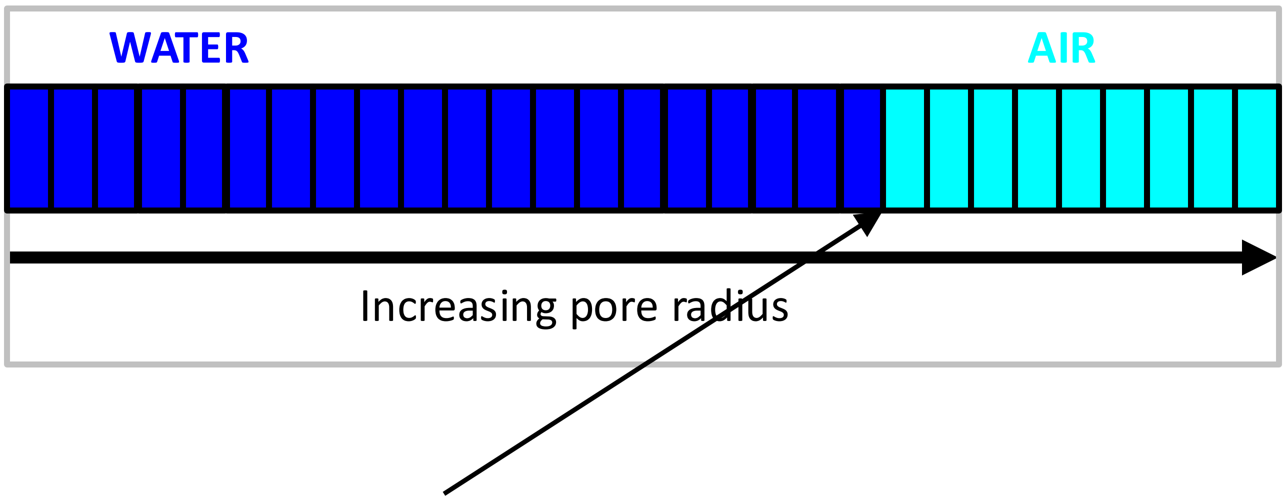
Capillary theory determines the largest pore radius that can retain water for a given matric potential (think suction)

Conceptual model of wetting/drying



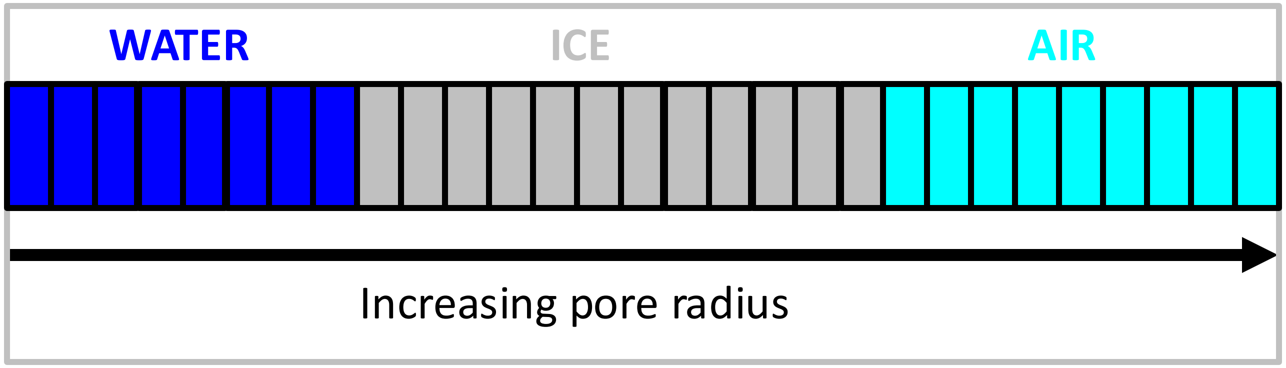
REWETTING

Conceptual model of wetting/drying



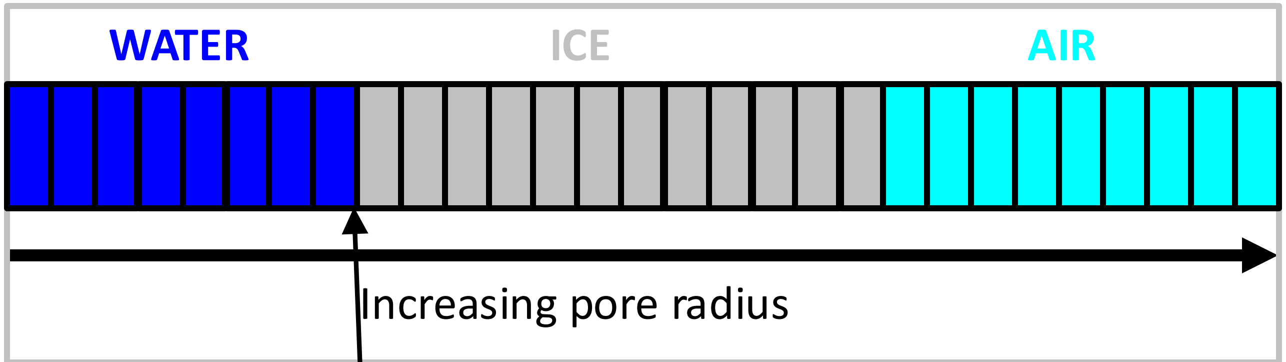
Capillary theory determines the largest pore radius that can retain water for a given matric potential (think suction)

Conceptual model of freezing soils



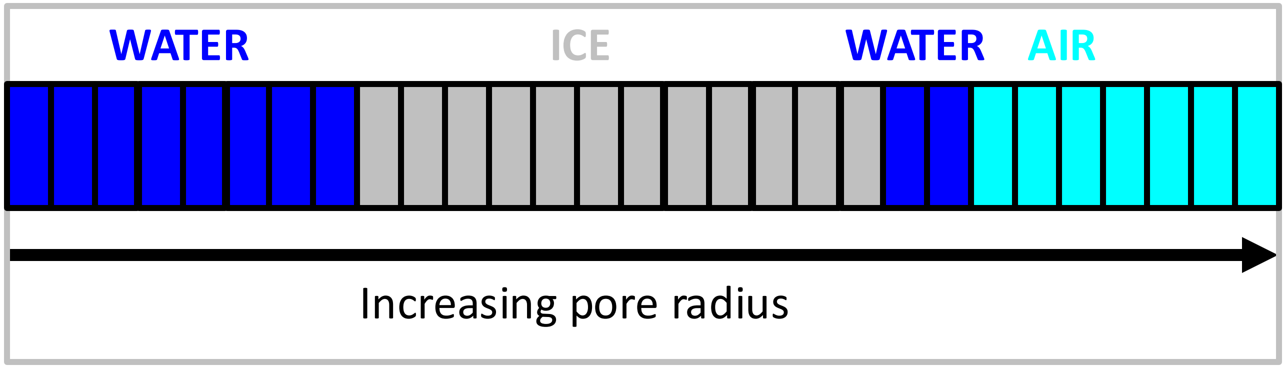
FREEZING

Conceptual model of freezing soils



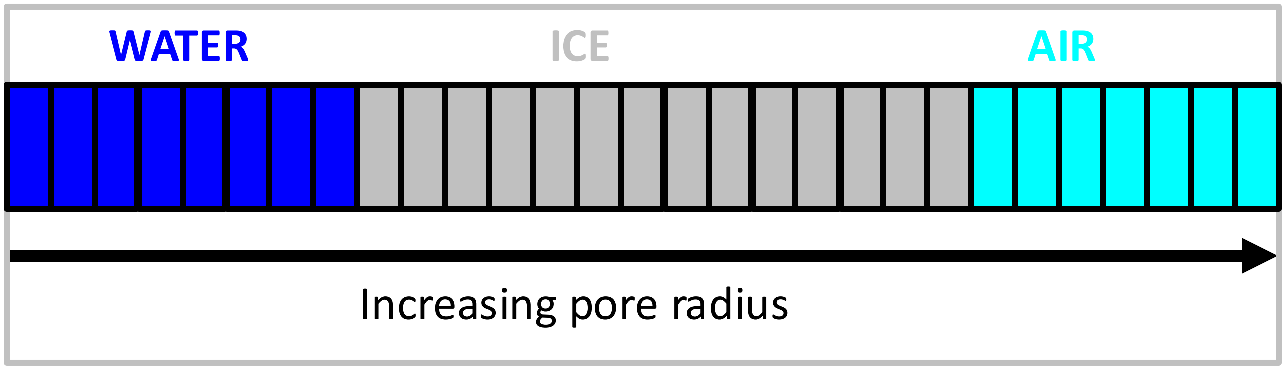
Water in smaller pores is harder to freeze. For a given subzero temperature, this threshold is determined by the *Generalized Clapeyron Equation*

Conceptual model of freezing soils



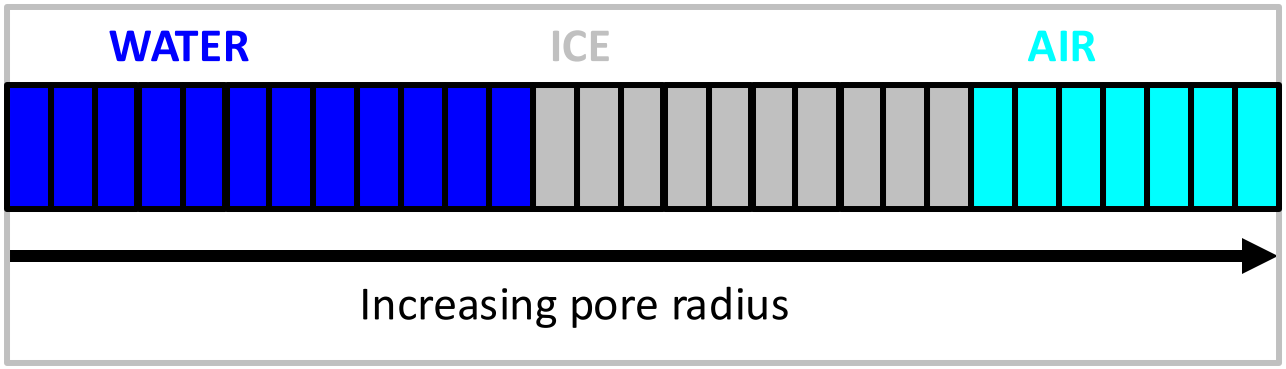
REWETTING

Conceptual model of freezing soils



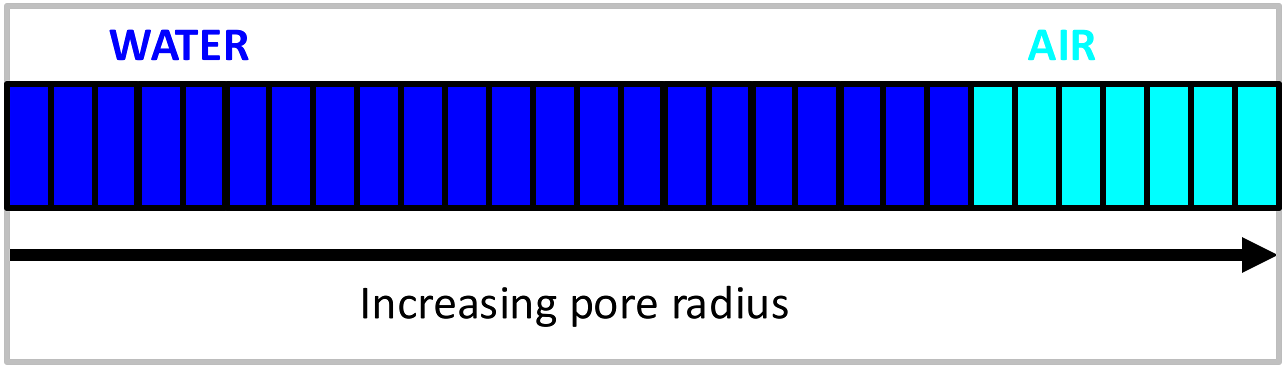
REFREEZING

Conceptual model of freezing soils



WARMING

Conceptual model of freezing soils



COMPLETE THAW

MODELING HYDROLOGICAL PROCESSES IN FROZEN SOILS

Putting it all together

The coupled mass and heat transport equations look something like this:

$$\overbrace{\boxed{\frac{\partial\theta_l(\psi)}{\partial t}}}^{\text{SMC}} + \overbrace{\boxed{\frac{\rho_i}{\rho_l}\frac{\partial\theta_i(T)}{\partial t}}}^{\text{SFC}} = \frac{\partial}{\partial z}\left[\overbrace{\boxed{K(\psi)}}^{K(\psi)}\left(\frac{\partial\psi}{\partial z}+1\right)\right] + \frac{1}{\rho_l}\frac{\partial q_v}{\partial z} - S$$

$$\underbrace{\boxed{C\frac{\partial T}{\partial t}}}_{\text{Sensible heat}} - \underbrace{\boxed{L_f\rho_i\frac{\partial\theta_i}{\partial t}}}_{\text{Latent heat}} = \frac{\partial}{\partial z}\underbrace{\boxed{\left[\kappa_T\frac{\partial T}{\partial z}\right]}}_{\text{Thermal conductivity}} - \rho_l c_l\frac{\partial q_l T}{\partial z}$$

In other words, these two equations incorporate pretty much everything we have discussed today (and some stuff we haven't – vapor flow!)

These equations are solvable. The problem is **quantifying the hydraulic properties** under different temperature/moisture conditions. This is a major scientific challenge